



School of Computer Sciences

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CDS505 DATA VISUALISATION & VISUAL ANALYTICS

Group 02

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Contents

1. Introduction and background.....	2
1.1. Website Link.....	2
1.2. Problem Overview.....	2
1.3. Dataset Description & Preparation.....	2
1.4. Target Audience.....	3
1.5. Related Work.....	3
2. Data Abstraction.....	3
2.1. Dataset Types.....	3
2.2. Data Types.....	4
3. Task Abstraction.....	5
4. Design Justification.....	13
4.1. Visual Encoding and Analytics(Marks & Channels).....	13
5. Four Levels of Validation.....	17
5.1. Domain Validation.....	17
5.2. Data Validation.....	17
5.3. Design Validation.....	17
5.4. Algorithm Validation.....	18
6. Final Visualisations.....	19
7. Discussion/Reflection.....	22
8. References.....	23

1. Introduction and background

1.1. Website Link

[US Shopping Trends & Consumer Behavior Analysis](#)

1.2. Problem Overview

In today's fiercely competitive market, it is no longer feasible to make decisions without the nonsense of data analysis. The biggest problem facing merchants at present is that if they don't really understand customers, they can't sell things to the right people. Blind promotion is not only a waste of money, but also may push customers further.

We chose this "commercial sales data set" precisely because it can help us break the blind spot. This data not only tells us the details of the customer, such as age, gender, and location, but also records the customer's behavior data, such as do you want to subscribe to the membership. Do you like to use the discount code? Which season do you like to buy?

The core goal of this project is to reasonably analyze and show the behavior of these customers through data analysis from these seemingly messy numbers, so as to facilitate merchants to provide more accurate personalized services. Profits and costs can also be calculated more clearly.

1.3. Dataset Description & Preparation

The data set used in this project comes from the "Shopping Behavior and Preferences Study" of the Kaggle platform, which captures the diversity of consumers in the retail environment through about 3,900 detailed records. Purchasing behavior. The data set covers 18 core attributes, not only including basic demographic data such as age, gender and geographical location, but also deeply recording the details of a single transaction, such as purchased items, commodity categories, consumption amount (USD) and purchase frequency.

Change to The data set has 18 attributes such as age, gender and geographical location. It also records transaction information such as purchased items, commodity categories, consumption amount, etc., such as purchased items, commodity categories, consumption amount (US dollars) and purchase frequency. In addition, this data also includes the degree of subjective preferences and market feedback of users, including (color, size, seasonal selection, evaluation score, marketing strategy, personalized customization method, discount application, promotional code and logistics method, etc.). These multi-faceted information provide a lot of data support for screening accurate consumers, analyzing and filtering out high-value customer groups, and evaluating strategic decisions for promotional activities.

In order to ensure that the subsequent work and tasks can be accurately and efficiently analyzed in the visualization tool, the raw data before import is strictly pre-processed. First of all, we cleaned the data set, excluded some duplicate records through Customer ID, and verified the integrity of the data to ensure that there were no missing values, thus ensuring the accuracy of the analysis results. Secondly, in order to improve the effect of visual coding, we have converted some variables, such as: dividing consecutive ages into age groups (15-20, 21-30, 31-40, 41-50, 51-60 and 60+) in order to more clearly perform the population in the dashboard. Statistical comparison. Finally, according to the requirements of the course for data abstraction, we identified the attribute types of each field of standardized calibration, which can

correctly match: numerical fields (such as consumption amount) and category fields (such as commodity type) and their respective visual channels, for further rationalized design in the future (Design Justification) lay the foundation.

1.4. Target Audience

This analysis report is primarily intended for the following audiences:

Retail store managers: It aims to provide them with a clear view of market trends, enabling informed decisions on product selection and pricing.

Marketers: The report assists in identifying high-value customers and assessing the success of advertising and promotional efforts.

Data analysis enthusiasts: This includes students or researchers keen on understanding user behavior and creating business charts, much like myself.

Supply chain managers: They can use this information to organize logistics and distribution based on seasonal and regional sales forecasts.

1.5. Related Work

Before starting this project, we reviewed various materials and discovered a wealth of foundational research in this area. Traditional market analysis typically employs the RFM model (Recency, Frequency, and Monetary value) to assess customer worth and segment distinct consumer groups (Hughes, 1994a). Numerous studies have also explored how demographic factors, such as age and gender, significantly influence shopping behaviors (Solomon, 2020), as well as how seasonal trends act as critical external drivers affecting the sales of specific categories like clothing and electronics (Radas & Shugan, 2008). Additionally, with the ubiquity of online shopping, consumers increasingly rely on electronic word-of-mouth; consequently, measuring customer satisfaction through sentiment analysis of reviews has become a pivotal research methodology (Pang & Lee, 2008).

In terms of tools and methods, handling large-scale datasets with traditional spreadsheet software like Excel has become increasingly inefficient. To bridge the gap between complex data and decision-making, Tableau has emerged as a crucial business intelligence tool. By adhering to fundamental information visualization principles (Tufte, 2001), this project utilizes Tableau to create interactive dashboards. This approach aligns with the concept of "storytelling with data," transforming raw numbers into engaging narratives that facilitate intuitive understanding for non-technical stakeholders (Knaflic, 2015).

2. Data Abstraction

This section analyzes our dataset's data abstraction through the Munzner framework (Munzner, 2014). This section includes identify dataset types and identify data types.

2.1. Dataset Types

This dataset has three different types - table, Geometry and cluster.

1) Table

The original dataset is a table, it has 3900 rows and 18 columns. Each row is an individual customer transaction, they are the table's items. And each

column is the customers' attribute, they are the table's attributes such as age, gender and so on. This table helps to create the visualizations like bar charts, pie charts, box plots and heatmaps. This type can be used for visualization to compare the sales situations of different categories, etc.

2) Geometry

In this dataset, there is a column that stores the position. This dataset can be transformed to the geometry type through this position attribute. These positions as the spatial marks, can map to the geospatial regions of the United States. This type can create a map to visualize the purchase amount and customer density. Can distinguish different areas by the shade of the color, and transform the text to the spatial positions.

3) Cluster

The cluster is not an explicit type. In our visualization, it is shown by the scatter diagram and the heatmap. Present the relationship between product categories, colors, and seasons to visualize to the heatmap. Sales volume, number of customers, and regions are visualized to the scatter plots.

2.2. Data Types

According to the measurement scales and ordering properties, classify these attributes to three categories - Categorical, ordinal and quantitative.

1) Categorical

Attributes: Gender, category, item purchased, location, color, season, subscription status, shipping type, discount applied and payment method.

These attributes are used to distinguish different identities, and they do not have implicit ordering or magnitude (Munzner, 2014). For example, the cloth and footwear are different categories, and they do not have the greater mathematical relationship. In our visualizations, these attributes are used in different color hues to compare them.

2) Ordinal

Attributes: Size and frequency.

These attributes have the logical and intrinsic ranking. For example, size include S, M, L and XL. This is a type of sorting. And for the frequency of purchases, Weekly, Monthly, Annually also is a type of sorting. In visualization, these attributes must be sorted by rank to show the trends of sales.

3) Quantitative

Attributes: Age, purchase amount (USD), review rating and previous purchases.

These attributes have the mathematical attributes, and they can derive other hidden values through mathematical calculations. For example, the average of the customer and the purchase amount becomes a central mark. Through this mark can get the distribution of the points.

3. Task Abstraction

Task 1: Shopping Trends Overview by Location

Discover: By observing [Figure 1](#), users can find consumption preferences in different regions, identify which areas are core markets and which are cold areas, and mainly distinguished as follows:

Discovered Matric	Description
Geospatial Patterns	Users can find the difference in customer density of Customers Location which is the different States.
Demographic Profiles: Gender, Subscription Status, Discount Applied, Age	Discover the main audience is male or female. Discover the proportion of loyal users. Discover the extent to which users use the promotion. Discover the main consumption age group of consumption.
Behavioral Habits: Frequency of Purchase, Payment Method, Shipping Type	Discover the user's repurchase cycle. Discover the payment channels preferred by users. Discover the user's preferences for logistics.
Seasonality and Sentiment: Season of Purchase, Product Ratings	Discover the seasonal fluctuations in consumption. Discover the distribution of customer satisfaction.

Search:

	Target known	Target unknown
Location known	Want know the number of users of PayPal in Montana	Want to know the Payment prefer, without knowing which payment are in list.
Location unknown	Want know the performance of Texas in this dataset, without knowing its detail in data.	Find some special customer habits.

Identify:

Through the screening of each filter and the selection of different regions, users can directly identify specific values by hovering the mouse or reading the label directly under composite conditions.

Compare:

Trend comparison: bar chart (Age), pie chart (gender, SubscriptionStatus, Discount Applied)

Numerical comparison: horizontal bar chart (Frequency, Payment, Shipping, Ratings), logo (Season of Purchase), allowing users to quickly compare the subtle differences of different categories.

Geographical comparison: geographical location map, compare the gap by comparing the color shades between states.

Summarise:

By aggregating all the data, users first perceive the whole, and by integrating the customers' demographics in the geographical location pie chart on the map. As well as some different consumption habits, to distinguish different shopping trends in different regions.

Task 2: Shopping Trends Overview by Category

Discover: By observing the [Figure 2](#), users can find consumption preferences across different product categories, identify which items are best-sellers and which generate the most revenue, mainly distinguished as follows:

Discovered Matric	Description
Product Performance Patterns	Users can find the difference in Items broad and Product Categories (Accessories, Clothing, Footwear, Outerwear)
Accessories, Clothing, Footwear, Outerwear	The sales volume and sales revenue of a certain product under this subcategory(Blouse, Jewelry, Pants, Shirt, Dress, Sweater, Jacket, etc)
Demographic Profiles: Gender, Subscription Status, Discount Applied, Age	Discover the main audience is male or female. Discover the proportion of loyal users. Discover the extent to which users use the promotion. Discover the main consumption age group of consumption.
Behavioral Habits: Frequency of Purchase, Payment Method, Shipping Type	Discover the user's repurchase cycle. Discover the payment channels preferred by users. Discover the user's preferences for logistics.
Seasonality and Sentiment: Season of Purchase, Product Ratings	Discover the seasonal fluctuations in consumption. Discover the distribution of customer satisfaction.

Search:

	Target known	Target unknown
Location known	Want to know the specific revenue of Blouse in the specific item list.	Want know the highest revenue of Product Category, without knowing which Category is in the list.

Location unknown	Want to know the performance of Outerwear in this dataset, without knowing where it is placed on the dashboard.	Checking if accessories sell better than clothing
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Identify:

Through conducting screening and filtering, based on the (Frequency of Purchase, Payment Method, shipping Type, Season of Purchase, Product Ratings), to identify the sales situation of a certain type of product under specific conditions.

Compare:

Trend comparison: bar chart (Age), pie chart (gender, SubscriptionStatus, Discount Applied).

Numerical comparison: horizontal bar chart (Specific Items, Frequency, Payment, Shipping, Ratings), Icons (Product Categories), allowing users to quickly compare the subtle differences of different categories.

Categorical comparison: Compare the performance gap between major product lines using the top summary section with the Icons of different categories.

Summarise:

By aggregating all the data and integrating the customers' demographics in the product category overview. As well as some different consumption habits, to distinguish different shopping trends in different product categories.

Task 3: Who's Buying? A Gender and Age Structure Analysis of Consumers

Discover: By observing the [Figure 3](#), users can find the demographic composition of the customers, identifying mainly gender groups and age structures, mainly as follows:

Discovered Matric	Description
Demographic Composition	Users can find the Ratio of Customer Gender.
Market Dominance	Users can find the Customer Age Group Ratio, identifying which age segments constitute to the core of the market. Discover that the customer base is predominantly Male Discover that the 60+ age group is the largest single age segment

Search:

	Target known	Target unknown
Location known	Want to know the specific percentage of Male customers in the Gender chart.	Want to know the Age Group chart to see the overall not for a specific age range.

Location unknown	Want to find that where is the 15-20 age group is represented in the visualization	Find which demographic group is the core force in consumption
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Identify:

Users can pinpoint exact demographic proportions by reading the percentage labels on each pie slice so that users can easily see the basic composition of the consumer base

Compare:

Intra-chart Comparison: According to the Age Group pie chart, divide the age into 6 groups: 15-20, 21-30, 31-40, 41-50, 51-60, and 60+, users can see the size of the slices to compare different generations.

Inter-category Comparison: There is a comparison between of the gender versus the the age distribution.

Summarise:

According to the analysis of the dashboard and the gender ratio analysis, it can be seen that the proportion of male customers is much higher than that of female customers, so it is recommended to expand the female consumer market. According to the age ratio analysis, it can be seen that teenagers are not very interested in products, and there is not much difference in other groups, which shows that the products are not targeted enough for different age groups. It can be suggested that products can be launched for different age groups and targeted marketing measures can be taken.

Task 4: Are Promotional Tactics Really Effective

Discover: By observing the [Figure 4](#), users can find the relationship between various marketing tactics and customer spending behavior, and identify which strategies are effective, mainly as follows:

Discovered Matric	Description
Promotional Effectiveness	Users can discover the relationships among various promotional strategies.
Discount Influence	Discover the impact of discounts on the average transaction price(Almost no difference)
Membership Value	Users can find the difference between Subscribers and non-subscribers to evaluate the impact of the subscription model. (Almost no difference)
Loyalty Spending Distribution	Discover the median and variability of Purchase Amounts across different ordering, identifying the impact of different loyal customers. (Almost no difference)
Logistics Influence	Discover the average order values difference between shipping types.

	(Almost no difference)
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Search:

	Target known	Target unknown
Location known	Want to know the average ordering amount of Free Shipping in the Shipping Type	Want to know the Customer Loyalty spending not for a specific consumption count range.
Location unknown	Want to find the Discount impact analysis.	Find differences among high loyalty customers.

Identify:

Users can determine specific performance indicators by observing the height of the bar or through the midline, so as to understand the consumption trend of a loyalty group.

Compare:

Binary Comparison: Compare discounts and subscriptions with 'yes or no' intuitively to evaluate the effectiveness of these projects.

Category Comparison: The Shipping Type chart allows users to compare multiple logistics options

Summarise:

For the comparison of the unit price of discount customers, it can be seen that the discount is not very attractive to users. It can be recommended to use a step-by-step discount. The more the total purchase, the greater the discount.

The box chart of the number of repurchases and the customer unit price reveals the comparison of the customer unit price of new and old customers. It can be seen that there is no obvious difference in the customer unit price of new and old customers. It can be suggested that more preferential measures should be taken for old users.

For the impact of the membership system, it can be seen that there is no obvious difference between membership or not, which shows that the membership system has not achieved good results. It is recommended to optimize the membership system, points, etc.

For the impact of the logistics mode, there is no difference, which shows that there is no particularly big difference between logistics. You can cooperate with logistics companies to win discounts and increase discounts.

Task 5: What Influenced Our Review Rating

Discover: By observing the [Figure 5](#), users can find key factors influencing customer satisfaction, identify which product attributes or operational services make the ratings higher, mainly as follows:

Discovered Matric	Description
Satisfaction Drivers	Users can find the difference in Average Review Rating across broad dimensions like Category, Gender, Shipping, and Subscription.
Product Attributes: Category, Color, Specific Items	Discover which product lines receive the best feedback. (clothing is the lowest) Discover if specific colors are preferred in certain contexts. (yellow in spring, grows in winter, has more consumption) Discover which individual items are customer favorites. (sandal highest, jeans lowest)
Operational Service: Shipping Type, Subscription Status	Discover if faster shipping correlates with higher ratings. (prefer high standard) Discover if loyal subscribers rate products more leniently or harshly. (Almost no impact)
Contextual Influence: Seasonality	Discover if satisfaction varies by Season for different categories or colors. (Spring accessories, autumn footwear, winter outerwear)

Search:

	Target known	Target unknown
Location known	Want to know the specific average rating of Sandals in the Item Purchased chart.	Want to see which colors generally perform best in Winter not for a specific color.
Location unknown	Want to find where the Winter season performance is analyzed on the dashboard.	Find some specific categories performing poorly in specific seasons.

Identify:

Users can determine the satisfaction score by viewing the digital label at the top of the bar chart or hovering the mouse over a specific cell in the heat map to see the exact score of a color product in a category.

Compare:

Stability Comparison: Evaluate different population groups and operating models through the bar chart at the top. It can be found that the scores are highly consistent regardless of gender, mode of transportation or subscription status (roughly between 3.7 and 3.8).

Pattern Comparison: The heat map can compare the dark blue area with the lighter area to identify specific seasonal advantages (outwear in winter).

Ranking Comparison: The bar chart can compare different items and distinguish the items with excellent performance such as sandals, the and items with low ratings such as jeans.

Summarise:

The bar chart above reveals the impact of different categories, genders, logistics and subscriptions on the evaluation score. There is not much difference in the score, but it can be concluded that footwear has the highest score, and for logistics, the standard weight is the highest.

For the heat map, the darker the color, the better the evaluation. It can be seen that Spring's Accessories, Outerwear and Fall's Footwear scores are the highest, and targeted improvement measures can be made.

For the scores of different Item Purchased, sandals have the highest score and jeans have the lowest score. Targeted improvements can be made to the 3 highest scores and 3 minimum scores.

Task 6: People Here Love Shopping

Discover: By observing [Figure 6](#), users can find the correlation between customer count and spending amount, which are mainly as follows:

Discovered Matric	Description
Market Correlation	Users can find if there is a positive or negative correlation between the number of Customers in a state and their Average Purchase Amount.
Regional Performance Quadrants	The quadrant scatter plot of customer unit prices in different regions. According to different quadrants, the more left the number of customers, the lower the number of customers, and the higher the customer unit price. This chart reveals the consumption capacity of different regions.

Search:

	Target known	Target unknown
Location known	Want to know the specific average purchase amount in Montana.	Want to scan the High Customer cluster area to generally see if their spending averages are higher or lower than the blue dots.
Location unknown	Want to find where Arizona is to see if it's a high or low spender.	want to find a special markets (low customer count but high spending power) without for a specific state.

Identify:

Users can view the exact coordinates of the number of customers by locating a specific point and reading it or by hovering the mouse over the point. Such as identifying that Pennsylvania has an average purchase amount of \$66.57.

Compare:

Benchmark Comparison: Visually compare each state with the central x-axis and y-axis (the origin of the four quadrants) to judge whether the trading volume of a certain position is "above average" or "below average".

Color Comparison: The colors are distinguished based on the average value of the order amount. Blue indicates values above the average, while red indicates values below the average.

Summarise:

The dashboard is mainly used for market positioning and describes the relationship between quantity and quality. It can be seen from the figure that market size does not necessarily predict consumption capacity. Arizona is lower than the average customer volume, but has a very high average customer unit price. California has a high customer volume, but the customer unit price is lower than the average.

Task 7: Do Our Customers Like Our Outerwear

Discover: This [Figure 7](#) mainly shows the comparison of total sales amounts of different product categories in different seasons, as follows:

Discovered Matric	Description
Seasonal Performance Category	Users can find the difference in Purchase Amount for specific categories (Outerwear, Footwear, Accessories, Clothing) within each Season (Summer, Winter, Spring, Fall).
Category Dominance	Discover which product category is the primary revenue in all seasons. (Clothing in all seasons)
Seasonal Demand Shifts	Discover the impact of different seasons on sales of different categories.(Sales of spring clothes will decrease)

Search:

	Target known	Target unknown
Location known	Want to know the specific purchase amount of Clothing" in the "Winter".	Want to look the Spring panel to see which category performed best not for a specific item name.
Location unknown	Want to find the performace Outerwear within the Winter group	Checking if footwear ever overtakes clothing in revenue.

Identify:

The specific income value can be determined by observing the height of each column on the Y-axis grid line (the income of spring clothing is about \$27,000)

Compare:

Seasonal Comparison: In a single seasonal period, the height of the bar chart can be compared to judge the popularity of various products. In colder months, clothing is more popular than coat products.

Summarise:

This dashboard season summarized sales data and found that despite seasonal fluctuations, clothing is the main source of income throughout the year, and "coats" have not become the dominant category during the winter. Therefore, need to optimize the Outerwear, in terms of marketing and product strength enhancement.

4. Design Justification

4.1. Visual Encoding and Analytics(Marks & Channels)

Task 1: Shopping Trends Overview by Location

[Figure 1](#)

Item: Sales

Data Attributes: Customers Location, Gender, Subscription Status, Age, Discount Applied, Product Ratings, Shipping Type, Season of Purchase, Payment Method, Frequency of Purchase.

Visual Mark: Point, Bar, Pie, Area

Visual Channel: Color,vertical position(length) , horizontal position(length) ,Pie size

Encoding Charts	Description
Location Map (Customers Location)	To visualize Customers Location and show the number of customers in different states by color.
Pie Charts (Customers Demographics)	To visualize the whole relationship in Customers Demographics and show the Gender, Subscription Status and Discount Applied.
Vertical Bar Char (Age)	To visualize Age And show different age groups (15 ~ 60+).
Horizontal Bar Charts (Frequency of Purchase,Payment Method,Shipping Type)	To visualize the ranking of classified data and display Frequency of Purchase, Payment Method and Shipping Type.
Graphy with Text	To visualize Season of Purchase. Use seasonal icons (flowers, sun, leaves, snowflakes) with percentage text to

	show the distribution with data.
Graphy With Bar	To visualize Product Ratings, use emotional icons (smiley face, crying face) and bar length to show the proportion of satisfied, neutral and dissatisfied.

Task 2: Shopping Trends Overview by Category

[Figure 2](#)

Item: Sales

Data Attributes: Product Categories, Gender, SubscriptionStatus, Age, Discount Applied, Product Ratings, Shipping Type, Season of Purchase, Payment Method, Frequency of Purchase.

Visual Mark: Point, Bar, Pie, Area.

Visual Channel: Color, vertical position(length), horizontal position(length), Pie size.

Encoding Charts	Description
Icons with Text (Product Categories)	To visualize Product Categories and show the item Purchased and Revenues of specific item in Product Categories by color and text number.
Pie Charts (Customers Demographics)	To visualize the whole relationship in Customers Demographics and show the Gender, Subscription Status and Discount Applied.
Vertical Bar Char (Age)	To visualize Age And show different age groups (15 ~ 60+).
Horizontal Bar Charts (Frequency of Purchase, Payment Method, Shipping Type)	To visualize the ranking of classified data and display Frequency of Purchase, Payment Method and Shipping Type.
Graphy with Text	To visualize Season of Purchase. Use seasonal icons (flowers, sun, leaves, snowflakes) with percentage text to show the distribution with data.
Graphy With Bar	To visualize Product Ratings, use emotional icons (smiley face, crying face) and bar length to show the proportion of satisfied, neutral and dissatisfied.

Task 3: Who's Buying? A Gender and Age Structure Analysis of Consumers

[Figure 3](#)

Item: Sales

Data Attributes: Gender, Age.

Visual Mark: Pie

Visual Channel: Color, Pie size.

Encoding Charts	Description
Pie Chart (Customer Gender Ratio) (Customer Age Group Ratio)	Used to visualize the relationship between Customer Gender Ratio and Customer Age Group Ratio. And show different genders (Male/Female) and different age groups (15 ~ 60+), and the percentage composition in the total customer group.

Task 4: Are Promotional Tactics Really Effective

[Figure 4](#)

Item: Sales

Data Attributes: Average Purchase Amount, Purchase Amount, Status of Discount Applied Used, Customer Loyalty, Subscription Status, Logistics type.

Visual Mark: Bar

Visual Channel: Color, Vertical position, Horizontal position, Pie size.

Encoding Charts	Description
Bar Chart (Impact of Discounts) (Impact of subscription) (Impact of Shipping Type)	To visualize Impact of Discounts, Impact of Subscription and Impact of Shipping Type. Compare the average purchase amount under the classification (Yes/No) or multiple classification (The shipping methods) through the difference.
Box Chart (Impact of Previous Purchases)	To visualize Impact of Previous Purchases (Customer Loyalty). Show different loyalty groups (1-40) and the distribution of the purchase amount, including the median.

Task 5: What Influenced Our Review Rating

[Figure 5](#)

Item: Sales

Data Attributes: Average Purchase Amount, Purchase Amount, Average Review Rating, Product category, Gender, Logistics type, Subscription Status, Product items, Colors, Season.

Visual Mark: Bar

Visual Channel: Color, vertical position, horizontal position.

Encoding Charts	Description
Bar Charts (Review Rating by Category)	To visualize Review Rating by Category, Gender, Shipping, and Subscription. The

(Review Rating by Gender) (Review Rating by Shipping) (Review Rating by Subscription)	average score under different groups is shown through the height difference(No significant difference).
Heatmaps (Review Rating by Category and Season) (Review Rating by Category and Color)	To visualize Review Rating by Category and Season and Color and Season. Display the data through color shade blue to identify seasonal satisfaction patterns.
Sorted Bar Chart (Age)	To visualize Review Rating by Item Purchased. Display the scoring ranking of specific products from high to low.

Task 6: People Here Love Shopping

[Figure 6](#)

Item: Sales

Data Attributes: Average Purchase Amount, Purchase Amount, Number of Customers, Location Name(US State).

Visual Mark: Point

Visual Channel: Color, vertical position, horizontal position.

Encoding Charts	Description
Point Charts (Average Purchase Amount by Location)	To visualize the relationship between Average Purchase Amount Y position and Customers Xposition. Each point represents a state, and its position shows its performance in two dimensions. And, the shade of the color also represents the Average Purchase Amount.

Task 7: Do Our Customers Like Our Outerwear

[Figure 7](#)

Item: Sales

Data Attributes: Purchase Amount, Season, Product Category.

Visual Mark: Bar

Visual Channel: Color, vertical position.

Encoding Charts	Description
Bar Charts (Purchase Amount by Season and category)	To visualize Purchase Amount by Season and Category. And use different colors to distinguish

	Seasons, The height of the bars represents the Purchase Amount
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5. Four Levels of Validation

This section uses the Munzner visualization model to evaluate our visualization design. This section includes domain validation, data/task validation, design validation and algorithm validation.

5.1. Domain Validation

Through this domain level, can validate the seasonality, customer segmentation and regional performance are important problems for retail managers. Bakhrun et al. believe that the data visualization is a very important tool to understand the consumer behavior and support strategic decision-making (Bakhrun et al., 2025). Inventory planner express track seasonal peaks can help to prevent inventory overstock and ensures that inventory matches the fluctuating demand (Inventory Planner, 2025). We classified the customers by attributes such as age and gender. This classification function was verified by the Matellio company, they think this is a good way to increase the targeted sales (Matellio Inc., 2024). Analyzing the geospatial also can help users to track the regional sales performance. Ginesys emphasized that tracking the performance of stores in each region is very important for optimizing the logistics and marketing (Ginesys, 2025).

5.2. Data Validation

We transform the original data to the specific dataset type, it is very helpful to achieve the visualization and domain goals.

Table validation

We validated the table type in the dataset. We divided the original data into three types such as categorical, quantitative and ordinal. This abstraction is necessary, because we need to conduct different visualizations for different domain issues. For example, the average purchase amount. Some visualizations cannot be achieved with the unprocessed data.

Geometry validation

We validated that it is correct to treat the position attribute as geometric data. Users can use the map to understand the differences in purchase amounts, consumer groups and other data among different regions.

Cluster validation

It is also very important to find clusters from the dataset. Cluster can let users have a comprehensive view of the overall distribution of the data. For example, the scatter plot enables users to intuitively find which regions have higher purchase amounts and customer than the average. This can help users quickly filter out regions that better meet their requirements.

5.3. Design Validation

This level aims to validate whether the visualizations are suitable.

Choropleth Map

We use the map to analyze the customers' location. This is the most intuitive way to show the spatial data. Users can find the higher customer density areas by observing the shade of the color.

Bar Charts

We chose the horizontal and vertical bar charts to compare the data such as product categories and payment methods. Because people are best at comparing lengths on the same scale. Users can quickly identify the top-selling product category.

Heatmap

We used the matrix view and applied color saturation to evaluate and score by category can season. This way can handle two classification dimensions at the same time, and users can identify the high scores or low points during the specific seasons.

Box Plots

The box plot can verify the distribution of purchase amounts. It can show the median and quartiles, and show the customer loyalty.

Color Consistency

We use a unified color scheme. General data displayed in blue, and other classifications such as gender are used the specific colors. The color consistency can reduce the user's cognitive load.

Chart classification

We categorize the charts according to logical themes. For example, we put all demographic charts together. This function can show the correlations among them.

Threat Addressed

We reduced the threat posed by invalid coding. For example, we avoid using variables with numerous categories in pie charts. Because it would be difficult to read.

5.4. Algorithm Validation**Efficiency**

We used Tableau to process the dataset and generate these visualizations. The Tableau is a modern business intelligence tool with very powerful functions, the required computing cost can be disregarded. These interactions are processed in real time with very little delay. So this algorithm is suitable for data of this scale.

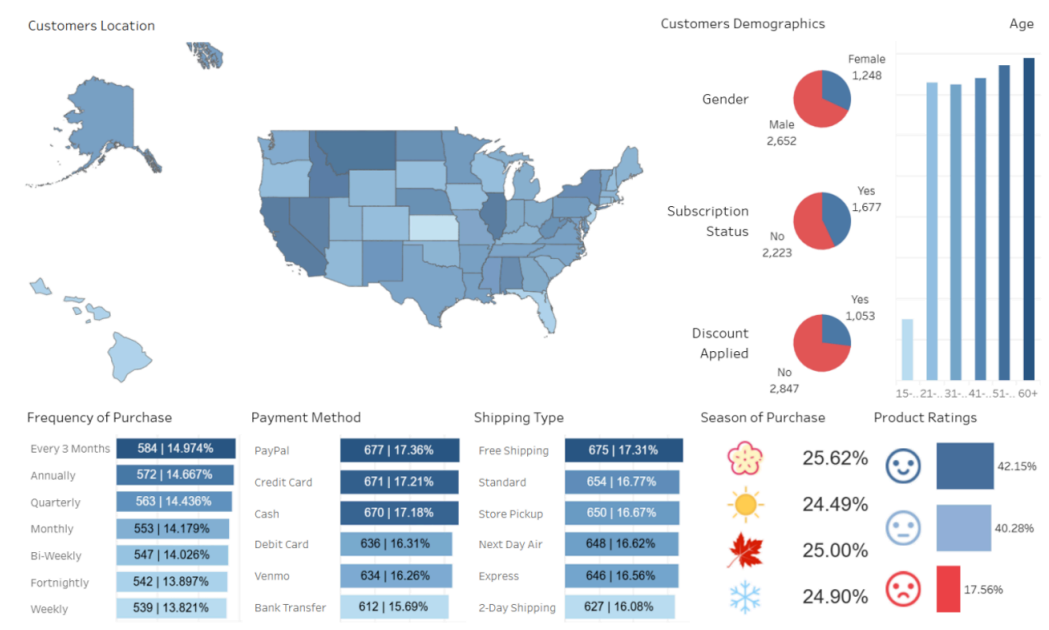
Data Integrity

We through clean data and remove blank data to ensure data integrity. This function can ensure the visualizations can display the content accurately, and avoid deviations caused by incomplete data.

6. Final Visualisations

Figure 1

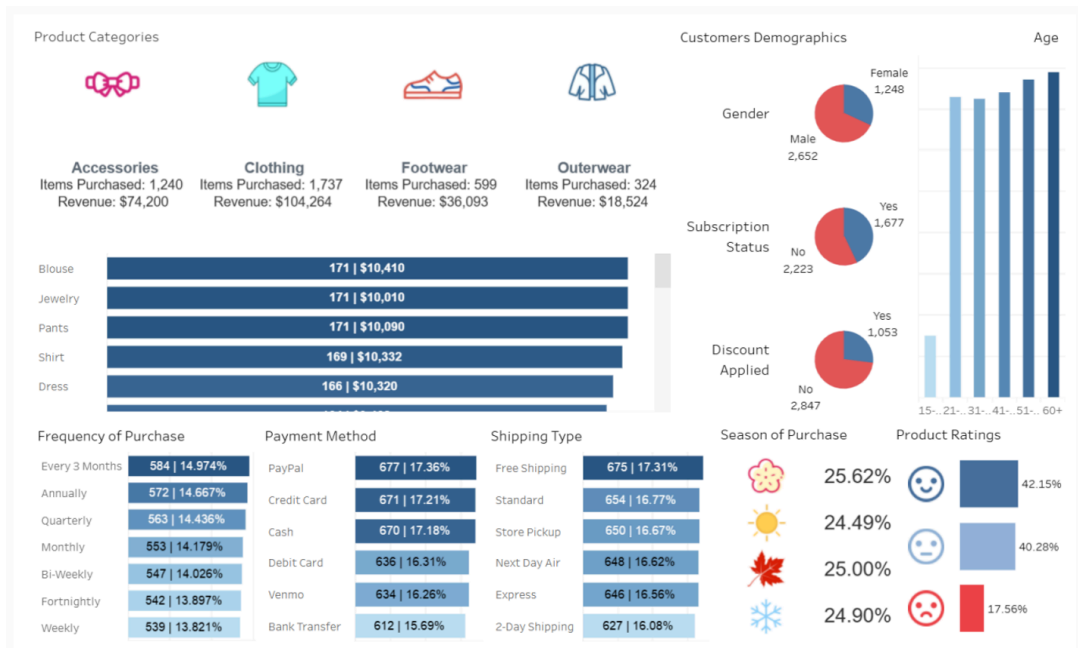
Shopping Trends Overview by Location



Note: Geographic distribution of shopping activity, regional preferences, and spending patterns across different states and urban/rural areas

Figure 2

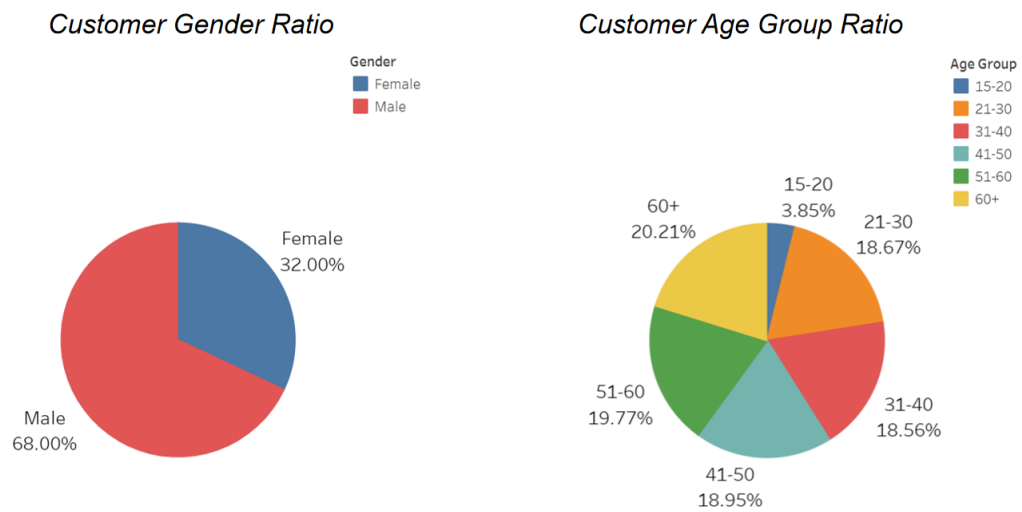
Shopping Trends Overview by Category



Note: Analysis of spending patterns across product categories, seasonal variations, and growth opportunities

Figure 3

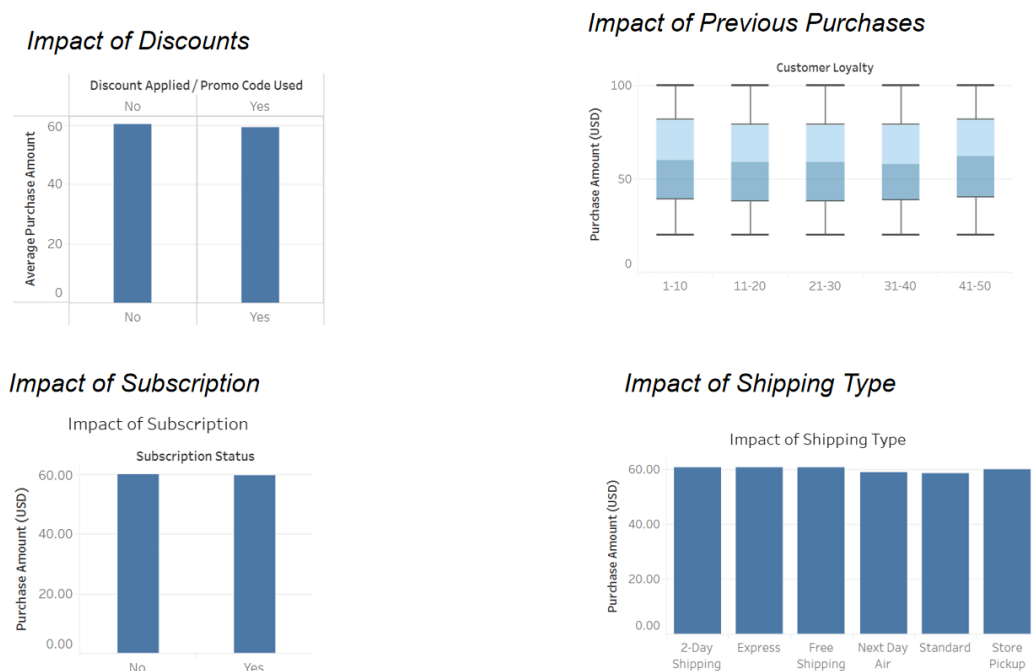
The distribution of customers by their gender and age



Note: Understanding the demographic composition of our customer base, purchasing power by age group, and gender-based preferences

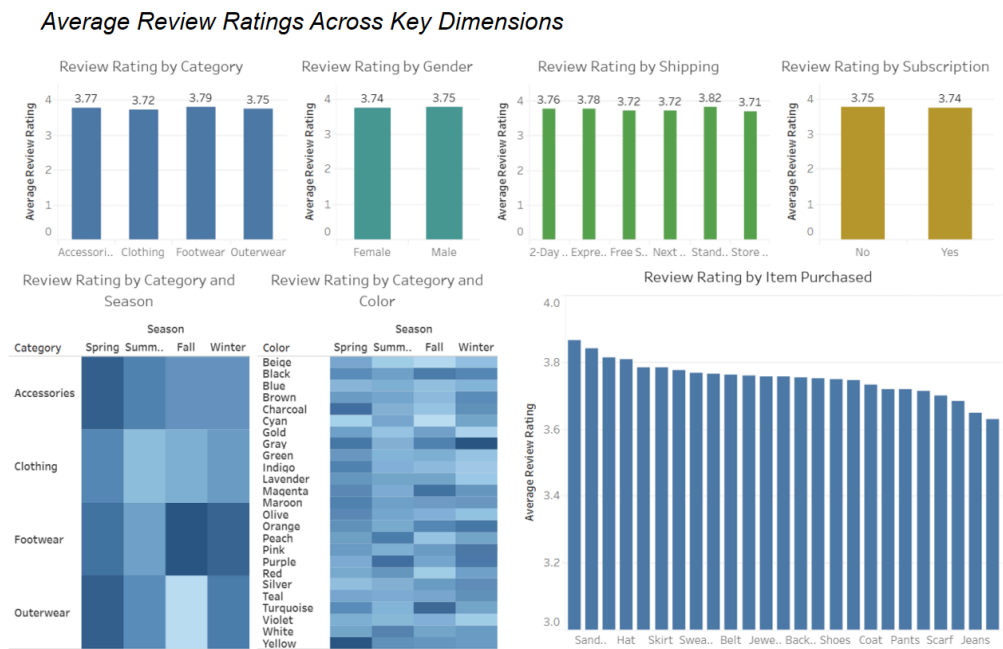
Figure 4

Impact of discounts, regular customers, subscriptions, and delivery methods on purchase unit prices



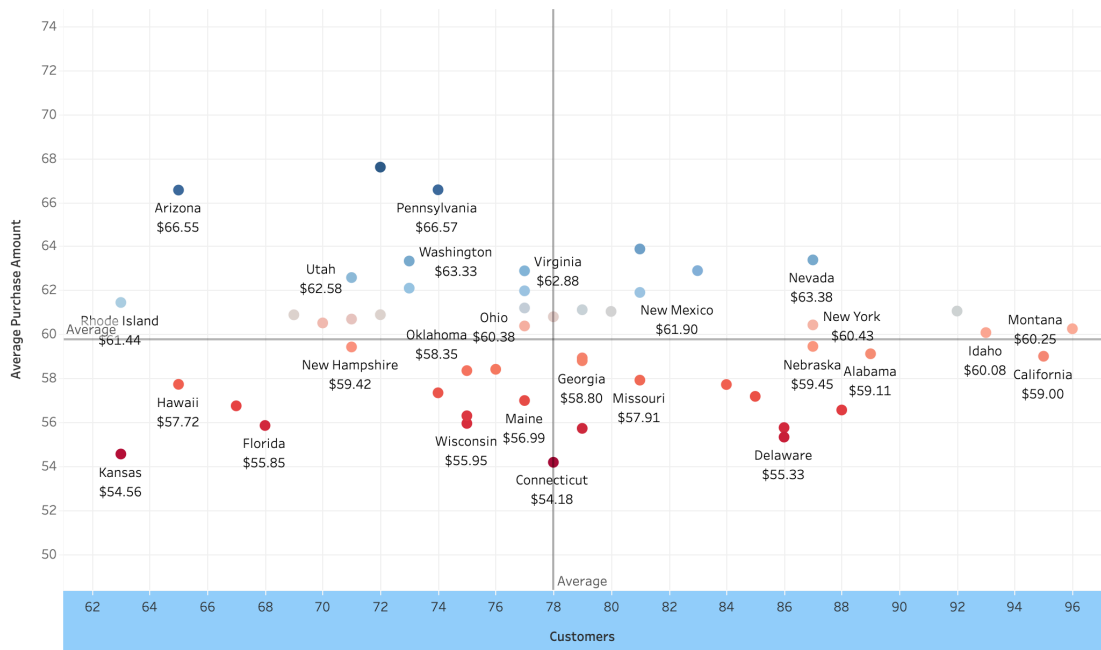
Note: Evaluation of different promotional strategies, their impact on sales volume, customer acquisition, and profit margins

Figure 5
What factors affect customer evaluations



Note: Analysis of customer review ratings, key factors influencing satisfaction scores, and correlation between ratings and repeat purchases

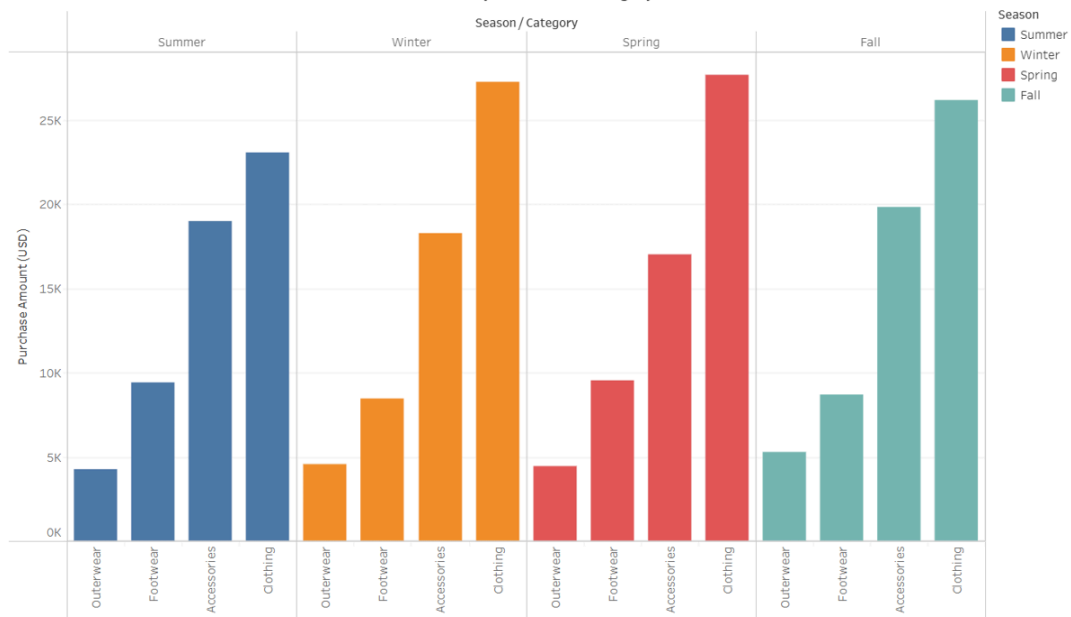
Figure 6
Average Purchase Amount by Location



Note: Understanding shopping frequency, channel preferences (online vs. offline), peak shopping times, and basket composition

Figure 7

Purchase Amount by Season and Category



Note: Deep dive into outerwear category performance, customer satisfaction, style preferences, and competitive positioning

7. Discussion/Reflection

When browsing various consumption reports, those macro trends will be disconnected from the daily life of users. Until the real shopping data set is opened, there are many stories behind everyone's consumption choices. This time, use the data of American consumers to try to play the role of "detective" and find out what factors are quietly affecting users' shopping carts.

Of course, data can't be used directly. The project spent a lot of energy on data cleaning, such as dealing with missing values, adjusting formatting inconsistency, etc. In order to make the follow-up analysis more intuitive, the age group and geographical location information is specially sorted out - after all, Tableau will be used for visualization later, and the clean and tidy data is the beginning of a wonderful story.

When sorting out data, imagine several different groups of people: if you open a store, you may be able to draw inspiration from product selection and pricing; if you do marketing, you may find out which promotion methods are really effective; even friends in charge of inventory management can use this method to predict the stocking of goods in different seasons. Of course, there are also curious bystanders who want to know why people pay.

Before starting to study, I learned how others did it. The classic RFM model is indeed commonly used. It mainly depends on the customer's recent purchase record, purchase frequency and consumption amount. In addition, there are many studies on how age and gender quietly affect our choices, or why people like to buy coats in autumn. Interestingly, it is now very common to comment when shopping online.

Many people began to judge whether the product was satisfactory by analyzing the emotions in the comments. These ideas will give a lot of inspiration.

Ultimately, data analysis tools are always changing. In the past, people relied on Excel, but now more and more people use R to process large amounts of data. For those who want to quickly "dialogue with data", tools like Tableau are much friendlier - this time I also followed this path: try to use Tableau to turn cold numbers into interactive and tellable visual stories.

8. References

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